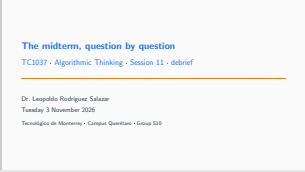
The midterm, question by question

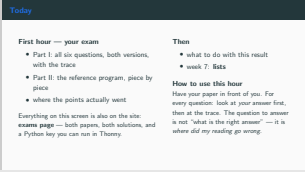


60 minutes on the exam, then 20 on what comes next. Hand back the papers first, face down, and say: do not look at the mark yet — look at the question while I explain it. Both versions are on every slide: **[A]** on the left, **[B]** on the right, same idea, different numbers.

2026-11-03

The midterm, question by question

Today


Today

First hour — your exam

- Part I: all six questions, both versions, with the trace
- Part II: the reference program, piece by piece
- where the points actually went

Everything on this screen is also on the site: **exams page** — both papers, both solutions, and a Python key you can run in Thonny.

Then

- what to do with this result
- week 7: **lists**

How to use this hour

Have your paper in front of you. For every question: look at *your* answer first, then at the trace. The question to answer is not "what is the right answer" — it is *where did my reading go wrong*.

Papers back, face down, before this slide. Ask them not to compare marks with the neighbour.

The midterm, question by question

How Part I and Part II were scored

How Part I and Part II were scored

Part I — 6 questions × 5 pts

All or nothing. Every line of the output, in the right order. A missing line, an extra line or a wrong order is 0 for that question — because a program that prints almost the right thing is wrong.

Part II — 70 pts, the academia's rubric

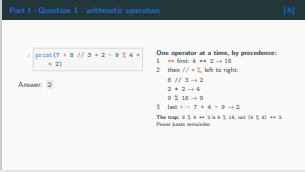
function with while	15
function with for	15
main()	3
main(): loop, validation, calls, labels	20
last line calls main()	1
technical requirements	−2 each

Each piece on the six-level scale: Excelente 100 % · Muy bien 85 · Regular 70 · Deficiente 50 · Muy deficiente 30 · Sin evidencia 0.

Say the all-or-nothing rule out loud: it is the single biggest source of “but I almost had it”.

The midterm, question by question

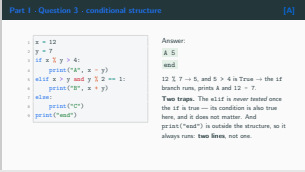
Part I · Question 1 · arithmetic operators [A]



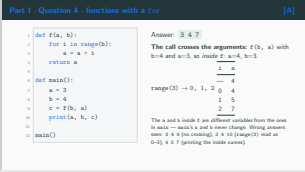
Write the ladder on the board as you go. Ask who got 1 or 7 — those come from doing % before **.

The midterm, question by question

Part I · Question 3 · conditional structure [A]



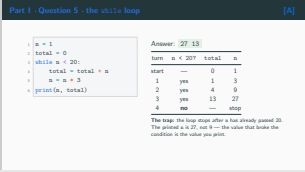
The most common loss here was writing only “A 5”. All or nothing: that is 0.



This was the designed trap of the paper. Spend two minutes here; draw the two boxes, main's and f's.

The midterm, question by question

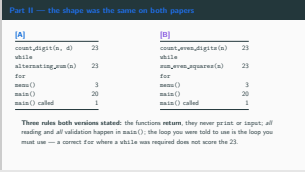
Part I · Question 5 · the `while` loop [A]



The table is the technique. Insist: one row per turn, columns for the condition and every variable.

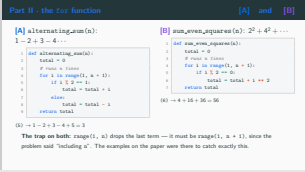
The midterm, question by question

Part II — the shape was the same on both papers



Say plainly: the single most expensive mistake was printing inside the function instead of returning.

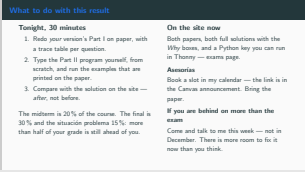
The midterm, question by question



Accept the alternative on A: total += (-1)**(i+1) * i. Ask if anyone wrote it that way.

The midterm, question by question

What to do with this result



What to do with this result

Tonight, 30 minutes

1. Redo your version's Part I on paper, with a trace table per question.
2. Type the Part II program yourself, from scratch, and run the examples that are printed on the paper.
3. Compare with the solution on the site — after, not before.

The midterm is 20 % of the course. The final is 30 % and the situación problema 15 %: more than half of your grade is still ahead of you.

On the site now

Both papers, both full solutions with the *Why* boxes, and a Python key you can run in Thonny — exams page.

Asesorías

Book a slot in my calendar — the link is in the Canvas announcement. Bring the paper.

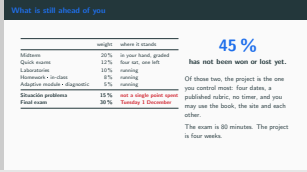
If you are behind on more than the exam

Come and talk to me this week — not in December. There is more room to fix it now than you think.

Pablo and anyone else deep in the hole: this is the invitation, said to the whole room so nobody is singled out. Then: next 20 minutes, lists.

The midterm, question by question

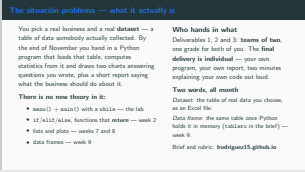
What is still ahead of you



Say the 45 % out loud and let it sit for two seconds. The point is not “you are in danger” — it is “there is more grade left than you have earned so far”. Then move straight to the project; do not dwell on the midterm again.

The midterm, question by question

└The situación problema — what it actually is



"No new theory" is the sentence that matters here — most of the fear of this project is fear of pandas, and they do not touch pandas until week 9. Teams of two: names on the board before they leave.

The midterm, question by question

The four deliveries — every date



The four deliveries — every date

	Due 23:59	What you hand in
1	Fri 6 Nov team of 2	An Excel table from a real, named source — a link, or a letter from the business (a family business counts). **30–50 rows**, up to 10 columns, about half text and half numbers, clean. Plus half a page: what the business is, what each column means. → `entregat1.xlsx`
2	Fri 13 Nov team of 2	Four to six *preguntas detonadoras* — **driving questions**: a manager really asks (*which branch sells most? which product?*), each answerable **from your own columns** and each naming the column and the statistic or chart that answers it. Two or three per person. → `entregat2.docx`
3	Fri 20 Nov team of 2	A program that loads `entregat1.xlsx` into a data frame and prints it — clean run, no errors, a function plus `main()`. The printed table matches the Excel file. → `entregat3.py`
F	Fri 27 Nov **individual**	The whole program, the report, and a two-minute oral explanation of your own code. → `entregaf_matricula.py` + report

All four: **Canvas → Modules → Situación problema**, never email. **Every stage missed or late costs 5 points** off the project grade.

Read the four dates out loud, in order, and have them write them down. The -5 per missed stage is from ACUERDOS 16 Sep — say it once, plainly, and do not repeat it.

The midterm, question by question

The final delivery — where its 100 points are



Program	50
extend(col1=col2+col3+1) — four or more statistics, tied to the question	12
graf(col) and graf(col1 — col2) — title, axis labels, legend	12
demo() / main() — the while, the if/elif/else, every input() is used	12
functions that **return**, a comment before each	9
main() — menu-driven, no crash, uses file	5
Report	30
Oral explanation — two minutes, no notes	20

The report, four things

1. a cover, the business, and what each column means
2. your driving questions **answered** — your numbers, screenshots of your charts
3. a conclusion: what the analysis shows
4. a **strategic proposal** — what the business should do about it

Why the oral exists. Pick any function and say what it does and why it is written that way. **A program you cannot explain is not yours**: under 10 caps the program at 25.

Two minutes on this frame, no more. The one thing to emphasise: the oral is not a trap, it is the reason you may use whatever tools you like — write code you can explain and there is nothing to fear.

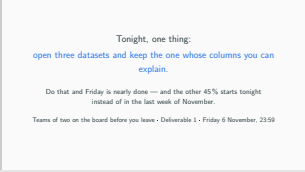


Leave this on screen while they photograph it. Then say the only two sentences that matter: the project has a deadline every Friday for four weeks, and the final exam is a Tuesday, not a Friday.



Say this to the whole room, looking at nobody. No names, no tone of disappointment — it is a procedure, read as a procedure. Item 4 is the one that buys the most points for the least work: say it twice.

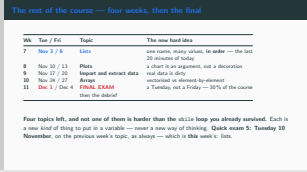
The midterm, question by question



Last slide. Do not add anything after it — the instruction is one sentence long on purpose. Collect the team names on the way out.

The midterm, question by question

└The rest of the course — four weeks, then the final



Read the four topics out loud and stop on the sentence at the bottom: nothing left is harder than the `while` loop. They do not believe that yet — say it anyway, because by week 10 they will.

Quick exam 5 is next Tuesday and it is on *lists*, i.e. on the twenty minutes that start now. Say the date; do not elaborate.

The midterm, question by question

Each of those weeks hands you the next piece of the project



Each of those weeks hands you the next piece of the project

Week	What it gives your program
7 lists	the columns of numbers your statistics run on — `estadisticasDescriptivas`
8 plots	`graficar` and `graficar2` — title, axis labels, legend
9 data	`entregar1.xlsx` loaded into a data frame → `entrega9.py`
10 arrays	a whole column at once instead of one row at a time

Look at the four deadlines again and then at this table: **every stage is due the Friday of the week that teaches it.**

Nothing in the project is taught *after* you need it — and nothing in it is taught twice. Miss a Tuesday and you are not behind on theory, you are behind on your own program.

The only piece you need *before* any of this is the one piece no lecture can give you: **the dataset**. That is why it is due first.

This is the frame that answers "why is there a deadline every Friday?". The answer is not discipline, it is construction: week 7 builds the statistics, week 8 the charts, week 9 the loading, week 10 the speed. Say the last line slowly — the dataset is the only part I cannot teach them.

The midterm, question by question

One name, one value — and then the dataset arrives

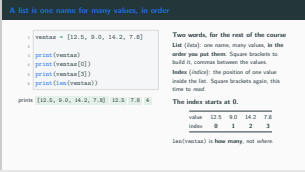


Put this on the screen and say nothing for five seconds — let them read 30 variable names. Then ask the room: “what happens when the business sends you one more row?” That question is the whole lecture.

Do not introduce the word *list* yet. Make them want it first.

The midterm, question by question

A list is one name for many values, in order

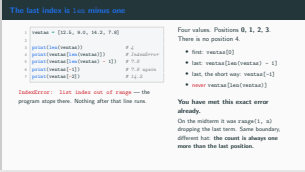


Write the value/index table on the board, by hand, and leave it up for the rest of the session — every error that follows is read off it.

Gloss both words in Spanish once: *lista* and *índice*. Then the sentence that matters: `ventas` is *one* variable, exactly like `total` was — it simply holds four things.

The midterm, question by question

The last index is `len` minus one



This is *the* frame for this group. Five of the six Part I questions and two of the five Part II losses were boundaries. Say it explicitly: "the mistake you made on Friday is the mistake this slide is about". Run the `IndexError` live in Thonny — they must see the red text at least once today.

The midterm, question by question

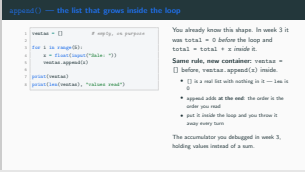
Two ways to walk a list — and when each is right



Both loops on the board, same output, and the question: which one can crash? Neither, if you write them like this — and that is the point of `range(len(...))`: `range` already respects the boundary for you. Mention the two-lists case; they will need it in week 8 for the charts (labels and heights).

The midterm, question by question

append() — the list that grows inside the loop



Ask where `ventas = []` must go, and let someone answer wrong first — putting it inside the loop is the classic bug and it prints a list of one value. Show that failure, then move the line.

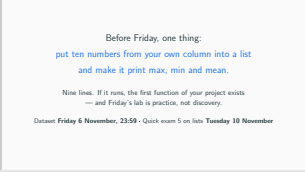
The midterm, question by question

Your first project function, on a list



Type this one live, from blank, in under three minutes — it is the payoff of the whole session: nine lines that do to 50 rows exactly what they do to 5. Then say the last sentence: this *is* the first 12 points of the final delivery, written five weeks early.

The midterm, question by question



Last slide of the session. Say the number: *nine lines*, because “write a program” sounds like an evening and nine lines sounds like ten minutes. Teams of two on the board on the way out.